UNIT 6 · TOPIC 1

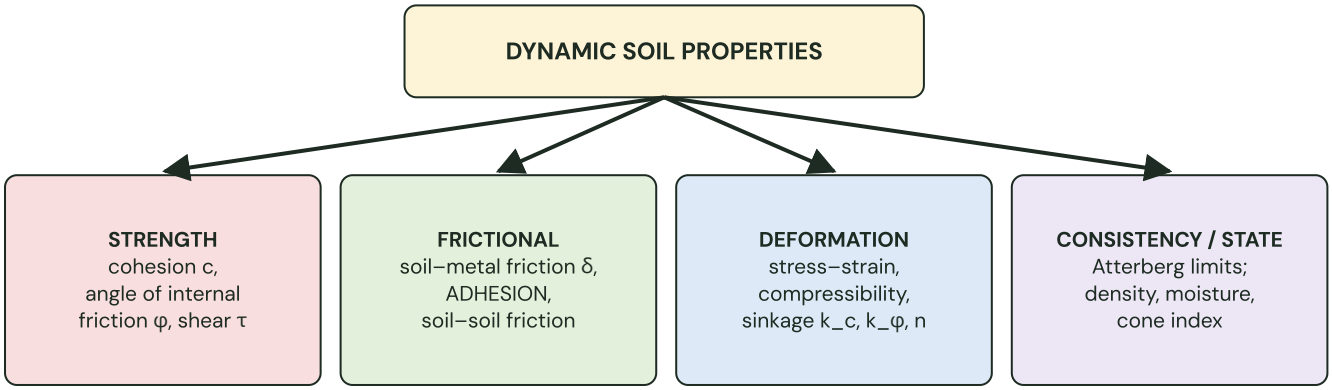
Dynamic Properties of Soil and Their Measurement

Syllabus topic: Dynamic properties of soil and their measurements

What soil dynamics is

- **Soil dynamics** studies soil behaviour — motion, deformation, failure — IN RELATION TO the forces applied by tillage tools and traction devices: soil as it is being WORKED, not the static soil of a soil survey.
- **Gill & Vanden Berg** (USDA Agriculture Handbook 316, 1967) defined the field and welded scattered results into a coherent body of knowledge. It underpins tillage-tool design (Topics 3–4) and traction (Topics 5–8).
- **Why "dynamic":** the same soil shows different strength and deformation depending on its CONDITION at the moment of loading — moisture, bulk density, rate of loading and confinement all change the response. A property measured while the soil is failing under a tool is dynamic; soil-survey descriptors are static.

The property groups



"Dynamic" = properties measured while the soil is being deformed/failed by a tool or running gear

(Gill & Vanden Berg, USDA Handbook 316, 1967) — distinct from static soil-survey properties

Dynamic soil property groups (redrawn from the source figure)

Group	Properties
STRENGTH	Shear strength τ, cohesion c, angle of internal friction φ (the Mohr–Coulomb pair)
FRICTIONAL / interface	Soil–metal friction angle δ and ADHESION (the "cohesion" analogue at the soil–metal interface — important for sticky black soils on mouldboards)
DEFORMATION	Stress–strain relationship, compressibility, pressure–sinkage parameters (Bekker k_c, k_φ, n — Topic 7)
CONSISTENCY / STATE	Atterberg limits (liquid, plastic, shrinkage), plasticity index, bulk density, moisture content, and the composite CONE INDEX

Plasticity index PI = liquid limit (LL) − plastic limit (PL)

Cohesion, internal friction and adhesion

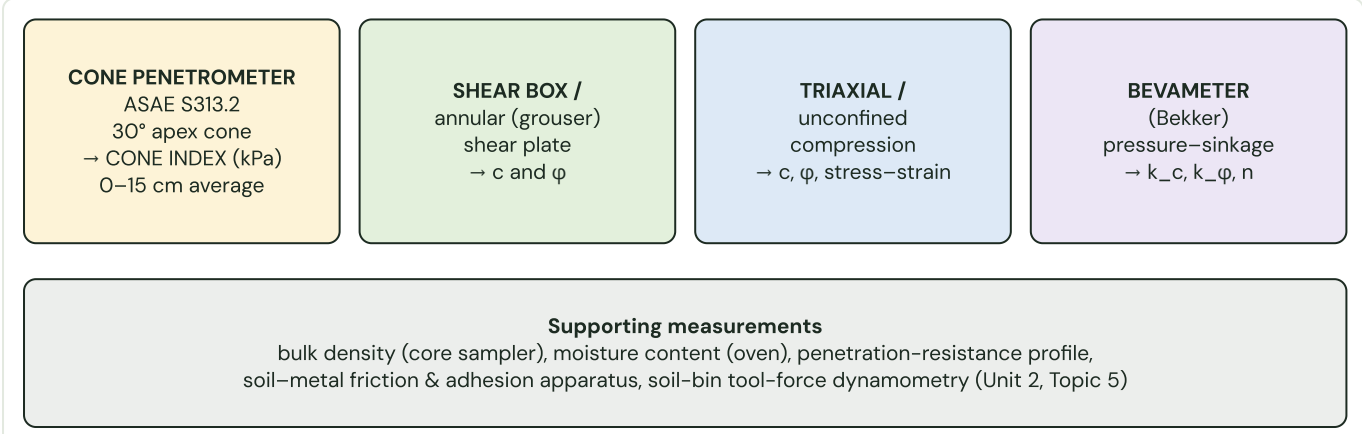
Property	Meaning	Typical behaviour
COHESION (c)	Shear resistance at ZERO normal stress — from inter-particle attraction and moisture films	High in clays; ~0 in dry sand
INTERNAL FRICTION (φ)	The frictional component that grows with normal stress	High in sand/gravel; lower in wet clay
ADHESION (c_a) and soil–metal friction angle (δ)	Soil sticking to, and sliding on, the tool surface	Govern drag and scouring of tillage surfaces

Teflon, enamel or polished surfaces reduce adhesion — important in Indian black cotton soil.

Cone index and the penetrometer (Liljedahl pp. 251–252)

- **CONE INDEX** = the average force per unit base area needed to push a cone-shaped probe into the soil at a steady rate (p. 251). Design and use are standardised in **ASAE S313.2** (**30° apex** cone).
- It is the practical single index of soil strength for TRACTION prediction (Topic 8).
- Cone index varies with depth. The **0–15 cm (0–6 in)** average correlates best for tyre sinkage **< 7.5 cm**; for deeper sinkage, average over the 15-cm layer that includes the maximum sinkage (p. 251).
- **Measure BEFORE traffic.** Freshly tilled (compactible) soils gain strength under load, so after-traffic cone index can be several times the before-traffic value (p. 252).

Measurement methods



Measurement of dynamic soil properties (redrawn from the source figure)

Instrument / test	Gives
Direct shear box; annular/grouser shear plate (in situ)	c and φ
Triaxial and unconfined-compression tests	c, φ and the full stress–strain curve
Bevameter (Bekker)	Pressure–sinkage k_c, k_φ, n and shear (Topic 7)
Soil-bin dynamometry	Tool forces (Unit 2, Topic 5)
Core sampler; oven drying	Bulk density; moisture content
Soil–metal friction/adhesion apparatus	δ and c_a

Practice MCQs

Q1. Soil dynamics is the study of soil behaviour in relation to:

(a) Soil classification
(b) Forces applied by tillage tools and traction devices
(c) Soil chemistry
(d) Rainfall

Q2. The foundational reference 'Soil Dynamics in Tillage and Traction' (USDA Handbook 316) was authored by:

(a) Kepner & Bainer
(b) Gill & Vanden Berg
(c) Goryachkin
(d) Bekker

Q3. A 'dynamic' soil property differs from a static one because it is measured:

(a) In a laboratory only
(b) While the soil is being deformed/failed
(c) After drying
(d) By remote sensing

Q17. Cohesion arises mainly from:

(a) Inter-particle attraction and moisture films
(b) Wheel load
(c) Air
(d) Temperature

Q18. Soil sticking to a tillage surface is governed by:

(a) Cohesion only
(b) Adhesion and soil–metal friction
(c) Internal friction only
(d) Cone index

Q19. Cone index is:

(a) A soil colour
(b) Force per unit base area to push a cone into soil at a steady rate
(c) Slip percentage
(d) Moisture ratio

Q20. The cone penetrometer is standardised in:

(a) ASAE S313.2
(b) ISO 500
(c) IS 4468
(d) SAE J833

Q21. The standard soil-cone apex angle is:

(a) 15°
(b) 30°
(c) 60°
(d) 90°

Q22. For tyre sinkage < 7.5 cm, the best-correlating cone index is the average over:

(a) 0–5 cm
(b) 0–15 cm (0–6 in)
(c) 0–30 cm
(d) 0–60 cm

Q23. Cone index for traction prediction should normally be measured:

(a) After heavy traffic
(b) Before the soil is subjected to wheel traffic
(c) After rain
(d) At night

Q24. In freshly-tilled compactible soil, after-traffic cone index compared with before-traffic is:

(a) Lower
(b) Same
(c) Several times higher
(d) Zero

Q29. Atterberg limits describe soil:

(a) Colour
(b) Consistency states (liquid, plastic, shrinkage limits)
(c) Strength directly
(d) Cone index

Q30. The plasticity index equals:

(a) LL - PL
(b) PL - SL
(c) LL + PL
(d) LL × PL

Q31. The Bekker bevameter measures:

(a) c only
(b) Pressure–sinkage parameters k_c, k_φ, n
(c) Cone index
(d) Moisture

Q32. Cohesion and internal friction can be obtained directly from a:

(a) Cone penetrometer
(b) Direct-shear / triaxial test
(c) Bomb calorimeter
(d) Patternator

Answers

Q1 — (b) Forces applied by tillage tools and traction devices
Q2 — (b) Gill & Vanden Berg
Q3 — (b) While the soil is being deformed/failed
Q17 — (a) Inter-particle attraction and moisture films
Q18 — (b) Adhesion and soil–metal friction
Q19 — (b) Force per unit base area to push a cone into soil at a steady rate — *Liljedahl p. 251 — cone-index definition.*
Q20 — (a) ASAE S313.2
Q21 — (b) 30°
Q22 — (b) 0–15 cm (0–6 in) — *p. 251 — 0–15 cm average for sinkage < 7.5 cm.*
Q23 — (b) Before the soil is subjected to wheel traffic — *p. 252 — measure before traffic.*
Q24 — (c) Several times higher — *p. 252 — compactible soils gain strength under traffic.*
Q29 — (b) Consistency states (liquid, plastic, shrinkage limits)
Q30 — (a) LL - PL — *PI = LL - PL.*
Q31 — (b) Pressure–sinkage parameters k_c, k_φ, n
Q32 — (b) Direct-shear / triaxial test

Sources: Gill, W. R. & Vanden Berg, G. E. — Soil Dynamics in Tillage and Traction, USDA Agriculture Handbook No. 316 (1967); Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 240–252; Kepner, Bainer & Barger, Ch. 6; ASAE S313.2 (cone penetrometer); classical soil mechanics (Coulomb 1776; Mohr); NPTEL soil-dynamics lectures.